

CCA GAME DEVELOPMENT

MAKING THINGS MOVE (X,Y)

Module 2 – Session 5

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Today's Goal

By the end of this session, you can:



**Understand X and Y
coordinates**



Change an object's position



**Make an object move with
code**



Control movement speed

How Does an Object Move?

Every game object has a position.

X



Left / Right

Y

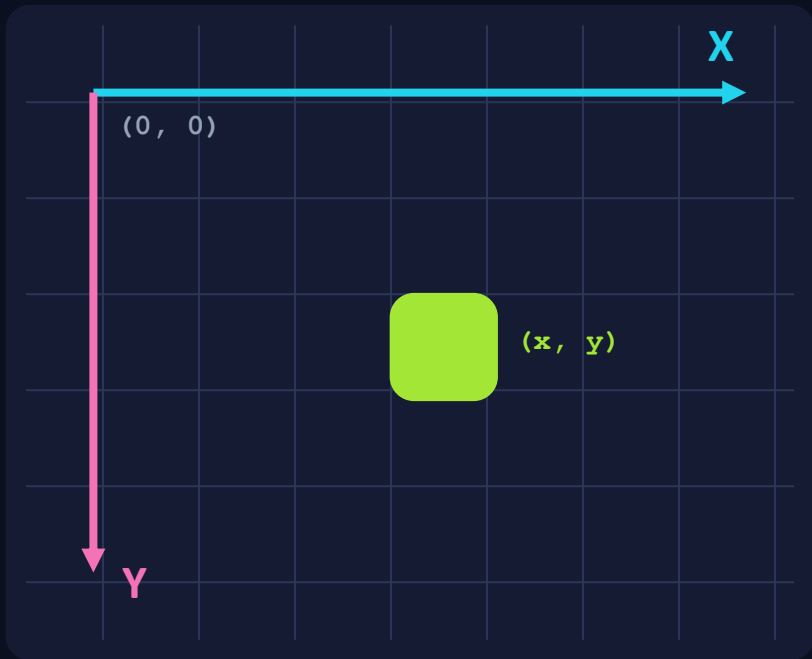


Up / Down

When we change the position, the object moves.

X & Y Coordinates

In Pygame



X increases → **RIGHT**



X decreases → **LEFT**



Y increases → **DOWN**



Y decreases → **UP**

Remember: X = Horizontal Y = Vertical

Position in Python

We can store an object's position in variables.



```
player_x = 100  
player_y = 200
```

```
Position = (100, 200)
```



`player_x`

→ horizontal position



`player_y`

→ vertical position

The Movement Formula

How do we make the object move?



In Python:



```
player_x += speed  
player_y += speed
```

X changes → Horizontal movement

Y changes → Vertical movement

What Does += Mean?

+= means "add to the current value"



```
player_x += 5
```



```
player_x = player_x + 5
```

is the same as

Example:

100



105



110



115



120

The value keeps increasing.

Move in Different Directions

CHANGE X



RIGHT

```
player_x += 5
```

CHANGE X



LEFT

```
player_x -= 5
```

CHANGE Y



DOWN

```
player_y += 5
```

CHANGE Y



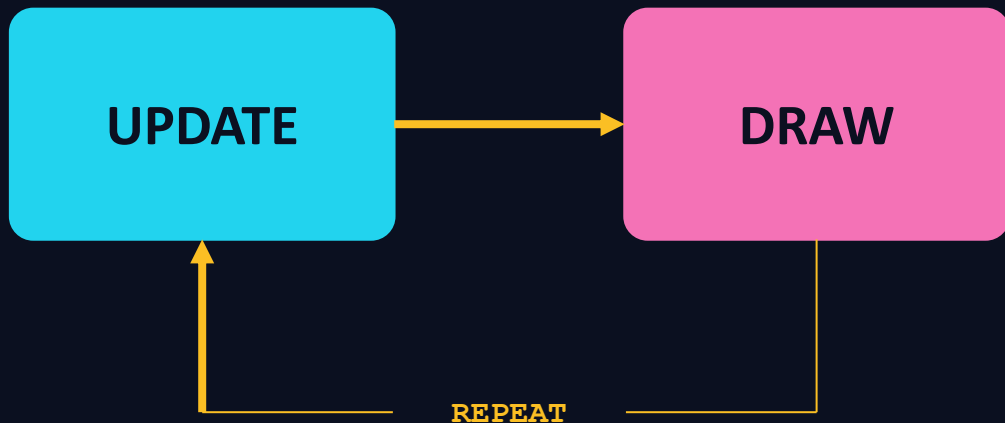
UP

```
player_y -= 5
```

X = Horizontal · Y = Vertical

The Game Loop

Why does the object keep moving?



The game keeps updating the object's position.

Repeated position changes make the object look like it is moving.

Let's Code!

Open the Session 5 starter code



01

Open the starter code

Use the Session 5 code from the class GitHub repository.



02

Find the variables

Look for `player_x` and `player_y`, then find the movement code.



03

Change, run, observe

Change the values, run the game, and observe the result.

<https://github.com/annisafzhn/CCA.git>

Control the Speed

What happens when we change the number?

`player_x += 1`



Slow

`player_x += 5`



Normal

`player_x += 10`



Faster

`player_x += 20`



Very fast

Bigger movement value → Faster movement

Challenges

1

CHALLENGE 1

Move LEFT

`player_x -= 5`



2

CHALLENGE 2

Move UP

`player_y -= 5`



3

CHALLENGE 3

Move DOWN

`player_y += 5`



4

CHALLENGE 4

Move UP + RIGHT

`diagonal` movement



Hint: Change X + Change Y → Diagonal movement

Try Your Own Movement

Create your own movement! Choose:



Direction

LEFT

RIGHT

UP

DOWN

Pick one



Speed

1

5

10

20

Pick one



Starting Position

```
player_x = ?  
player_y = ?
```

Choose your own X and Y values.

Today's Big Ideas



LEFT / RIGHT

Change X → Horizontal
movement



UP / DOWN

Change Y → Vertical movement



Game Loop

Repeats the movement

POSITION + CODE = MOVEMENT

NEXT SESSION

LISTENING FOR TRIGGERS



Keyboard Input



Player Control



Movement